

Michael Dias

London, Ontario, Canada

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Hands-on maker and Computer Science graduate with strong practical experience in embedded systems, real-time control, and rapid prototyping. Writes bare-metal drivers in C, interfaces sensors over I2C, SPI, and UART, tunes PID control loops against measured data, and isolates hardware faults from software faults at the bench. Builds complete systems end to end, from CAD and wiring through firmware to field testing. Seeking a first professional engineering role where that range is an asset and there are senior engineers to learn from.

Core Competencies

- **Embedded & Firmware:** C, C++, bare-metal development, STM32 (CubeIDE), ESP32, FreeRTOS, interrupts, timers, DMA, real-time debugging.
- **Sensor Fusion & Control:** Quaternion attitude estimation, 9-DOF Mahony filter, Kalman-filter-style correction, PID design and tuning, gyroscope noise filtering.
- **Interfaces & Protocols:** I2C, SPI, UART, CAN, BLE; writing sensor drivers from scratch without vendor abstraction layers.
- **Analog & Power Electronics:** Transistors, MOSFETs, buck regulators and switching supplies; fluent schematic reading and tracing; discrete component behaviour.
- **Hardware & Diagnostics:** Multimeter fault isolation, continuity testing, staged power-up, signal verification, soldering, crimping, PCB integration.
- **Industrial Controls:** PLC ladder logic, I/O addressing, tags, timers/counters, relay logic, Studio 5000 (certificate-level training; follow-on course in progress).
- **Software & Tooling:** Python, Flask, CMake, Git, Claude Code, unit testing (CTest), Linux/Windows/macOS, FreeCAD, 3D printing.

Engineering Projects

- **IMU Sensor Fusion & Real-Time Control** github.com/MichaelDias9
Personal Project 2025
 - Developed a 9-DOF Mahony filter using quaternions in C++ for real-time attitude estimation, with a RayLib front end for live 3D visualization.
 - Wrote I2C drivers in C from scratch for an MPU6500 accelerometer/gyroscope and an AK09911C magnetometer, building low-level hardware interfaces with no vendor abstraction layer.
 - Implemented configurable sensor rates and runtime PID tuning to evaluate control behaviour under different operating conditions.
 - Integrated third-party libraries across macOS, Linux, and Windows using CMake.
- **FPV Flight Controller / Drone Build** London, ON
Personal Project 2026
 - Matched motor KV rating, propeller pitch, ESC current rating, and Li-Po capacity to optimize thrust-to-weight ratio and electrical headroom.
 - Soldered high-current power systems and integrated flight controller, receiver, ESCs, and FPV video over UART.
 - Diagnosed integration and hardware faults through continuity testing, staged power-up, and signal verification before they escalated.
 - Tuned PID loops and gyroscope filtering from black-box flight logs, eliminating oscillation while preserving responsiveness.

- **Indoor Localization Mobile App** github.com/MichaelDias9
Startup Project 2024
 - Built a mobile app fusing GPS, pedometer, and BLE beacon data for localization in GPS-denied indoor environments.
 - Implemented beacon-based multilateration for absolute position measurements more accurate than GPS, applied as corrections in a Kalman-filter-style estimator.
 - Moved sensor fusion to a Python/Flask backend to sustain a higher processing rate than the handset allowed.
 - Recorded demonstration contributed to securing \$1,500 in early-stage funding.
- **Hardware Repair, Salvage & Reverse Engineering** **London, ON**
Ongoing 2023-Present
 - Diagnose and repair consumer and hobbyist electronics: isolated a non-charging FPV receiver/monitor to a failed internal LiPo cell rather than the charger or charging circuit, returning a written-off unit to service.
 - Salvage and repurpose components from decommissioned devices, identifying power requirements and IR remote pairing of undocumented boards for reuse in custom builds.
 - Self-taught analog electronics in parallel: transistors, MOSFETs, switching regulators, and schematic tracing.

Professional Experience

- **The Beer Store** **London, ON**
Warehouse Operations Associate 2021-Present
 - Operate electric ride-on pallet jacks in a safety-critical environment under strict PPE, SOP, and equipment-specific requirements.
 - Plan pallet builds and truck loading for efficient delivery routing and reliable material movement.
 - Resolve customer and delivery issues cross-functionally under time pressure while maintaining safety and service standards.

Education

- **Fanshawe College** **London, ON**
Programmable Logic Controllers – Follow-on Course (in progress) Sep–Oct 2026

Follow-up program to the PLC certificate below: ten 3-hour hands-on sessions (30 hours) building on ladder logic and Studio 5000 fundamentals. Expected completion October 2026.
- **Fanshawe College** **London, ON**
Certificate in Programmable Logic Controllers 2026

Ten weeks of hands-on industrial control training: ladder logic, tags, I/O addressing, timers, counters, industrial sequencing, relay logic, and Studio 5000.
- **Western University** **London, ON**
B.Sc. Specialization in Computer Science 2019-2023

Dean's Honour List throughout; 87%+ final-year average. Coursework in software design, databases, distributed systems, artificial intelligence, and systems programming.

Languages

- English (native).

Honors and Awards

- **Dean's Honour List** – Western University (2019–2023).
- **\$1,500 early-stage funding** – secured for a startup project following a product demonstration (2024).

References

- Available upon request.